

Pragya Sekar

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Executive Summary

Motivated Computer Science student with strong foundation in machine learning, cybersecurity, and cloud computing. Eager to apply technical and creative skills to solve real-world problems, contribute to teams, and grow as a builder and leader.

Education

Vellore Institute of Technology (VIT) Chennai

Aug 2023 – July 2027

B.Tech, Computer Science and Engineering

CGPA: 9.26/10

Key Courses: Data Structures & Algorithms, Operating Systems, Computer Networks, Theory of Computation, Cloud Computing, Database Systems, Cryptography and Network Security.

Experience

CPP Intern, Hewlett Packard Enterprise (HPE)

Feb 2026 – Present

Developing QueryCraft, an enterprise Text-to-SQL system allowing analysts to query HPE NonStop server performance data using natural language. Engineered the end-to-end pipeline using FastAPI, React, PostgreSQL, and the Google Gemini API. Implemented semantic caching with ChromaDB to reduce API costs, and enforced strict database security using SQLGlot AST validation for read-only query execution.

CV and AI Dataset Intern, Ecomotive Smart Solutions

Dec 2025 – Feb 2026

Developed a custom, representative image dataset of Indian municipal solid waste, sourcing and categorizing data across 5 distinct classes (hard plastics, soft plastics, bottles, cans, and general waste). Trained and optimized a computer vision object detection model on the custom dataset to automate and improve the accuracy of localized waste segregation pipelines.

Co-founder & Technical Member, vehnicate

Dec 2023 – Present

Technical member at vehnicate, a startup dedicated to bringing structure to the driving experience and road infrastructure through a highly interactive gamified ecosystem. Worked on object detection models for unmarked speedbumps, potholes, unpaved roads etc. Currently working on deploying on cloud (AWS)

ML Instructor, Workshop

June 2025

Co-led a 2.5-hour hands-on ML workshop covering FFT, SMOTE, and statistical techniques. Mentored students using real datasets with Python and Google Colab.

Projects

QueryCraft – Natural Language to SQL Performance Report Generator

2026

FastAPI, React, PostgreSQL, Google Gemini API, ChromaDB, SQLGlot, Python, JavaScript

- Engineered an enterprise-grade Text-to-SQL GenAI platform enabling natural language querying of complex HPE NonStop server telemetry, efficiently processing over 212,000 rows across 9 relational tables to autonomously generate dynamic visualizations and downloadable PDF/Excel reports.
- Architected a secure LLM pipeline integrating ChromaDB for semantic caching (achieving sub-second latency for frequent queries) and SQLGlot for Abstract Syntax Tree (AST) validation, ensuring strict read-only execution safety and robust error recovery through automated prompt retry loops.

TEG-NeSyNet – Neurosymbolic Knowledge Tracing for Educational AI

2025 [🏆]

PyTorch, PyTorch Geometric, Graph Neural Networks, Symbolic Reasoning, Python

- Developed a novel neurosymbolic AI framework combining heterogeneous temporal Graph Neural Networks (10.4M parameters) with educational psychology principles for student knowledge tracing, achieving **81.42% accuracy** and **91.32% AUC** on COVID-era ASSISTments dataset—outperforming state-of-the-art BKT methods by **5.72 percentage points**.
- Architected multi-head attention mechanisms to model complex student-problem-skill relationships while integrating a symbolic reasoning engine with 8 educational psychology rules using learnable weights, representing

the first neurosymbolic approach applied to knowledge tracing and positioning for publication at top-tier AI conferences.

MIDSight – Modular Linux Security CLI

2025 [🔒]

Python, AI/ML, Security

- Developed MIDSight, a modular Linux security monitoring CLI tool featuring process and network monitoring, file integrity checks, GeoIP threat tracking, honeypots, and automated reporting.
- Integrated Google Gemini AI to provide intelligent, natural language analysis of security events, built securely with Python 3.9+ and secure API key management.

Finesse – AI Document Intelligence Platform

2025 [🔒]

Node.js, React, Google Gemini, Qdrant, HuggingFace

- Developed a full-stack AI-powered document analysis platform enabling multi-format file processing (PDF, DOCX, images) with semantic search capabilities and real-time conversational querying through Google Gemini integration. Implemented vector database architecture using Qdrant for semantic search with HuggingFace embeddings, achieving **sub-3 second response times** and enterprise-grade security with no permanent data storage.

Road Infrastructure Detection System - vehnicate

2024-2025 [🔒]

Python, YOLO, OpenCV, FastAPI, HRNet, Computer Vision

- Developed a comprehensive computer vision system using custom YOLOv8 models to detect road defects including potholes, longitudinal cracks, unpaved roads, and marked speedbumps with integrated quality assessment algorithms for each detected object type.
- Implemented hybrid detection approach combining YOLO-based object detection with traditional computer vision techniques (Canny edge detection, Hough transforms, HRNet) for unmarked speedbump detection, deployed via FastAPI for real-time road infrastructure monitoring and analysis.

phishPicket – Phishing URL Detection

2025 [🔒]

Python, Pytorch, Flask, FastAPI

- Developed phishPicket, a machine learning-powered phishing detection system that analyzes URL structure, domain characteristics, and SSL certificate information to classify URLs as malicious or legitimate with **90% accuracy**. Built a comprehensive web application using Flask and FastAPI for real-time URL scanning, featuring automated threat assessment and instant security alerts for users.

VPS Hosting Platform Using OpenStack

2025

OpenStack, Linux, KVM, Python, Shell

- Architected and deployed a comprehensive VPS hosting platform leveraging OpenStack's Nova, Neutron, and Keystone services for automated virtual machine provisioning, network management, and identity authentication. Implemented KVM hypervisor integration with Python automation scripts and shell scripting for seamless VM lifecycle management, including dynamic resource allocation, monitoring dashboards, and multi-tenant user access controls for scalable cloud infrastructure deployment.

Technical Skills

Programming Languages: Python, C++, Java, JavaScript, C

Web Technologies: React, Node.js, HTML5, CSS3, Tailwindcss, RESTful APIs

Databases: MongoDB, MySQL, Qdrant (Vector DB)

ML/AI Frameworks: PyTorch, TensorFlow/Keras, scikit-learn, OpenCV, HuggingFace

Cloud & Infrastructure: AWS, GCP, OpenStack, Docker, KVM

Tools & Technologies: Git, FastAPI, Flask, YOLO, Linux

Additional Information

Runner-up – CUMTA Urban Move Challenge 2025, Mapping & Auditing Track (Theme 4), for Street Infrastructure Mapping & Auditing via Mobile App

Certifications: Artificial Intelligence with Machine Learning (PreGrad), MERN Stack (Ethnus)

Soft Skills: Problem-solving, Communication, Initiative, Teamwork, Creative Thinking